0580_s24_ms_22_Q24

Question	Answer	Mark	Partial Marks
24(a)	$\frac{1}{2}\mathbf{b} - \frac{2}{3}\mathbf{a}$	**2**	**B**1 for answer $\frac{1}{2}\mathbf{b} + k\mathbf{a}$ or $j\mathbf{b} - \frac{2}{3}\mathbf{a}$ or correct unsimplified in terms of **a** and **b**
24(b)	$\frac{5}{4}\mathbf{b}$	**3**	**M2** for $\overrightarrow{RS} = \frac{1}{4}\mathbf{b}$ oe or $\overrightarrow{MS} = \frac{3}{2}\left(\frac{1}{2}\mathbf{b} - \frac{2}{3}\boldsymbol{a}\right)$ oe or $\overrightarrow{NS} = \frac{1}{2}\left(\frac{1}{2}\mathbf{b} - \frac{2}{3}\boldsymbol{a}\right)$ oe or **M1** for a correct route in terms of vertices a nd/or **a** and/or **b** or **B**1 for answer $j\mathbf{b}$ where $j > 1$ or $\overrightarrow{RS} = \frac{1}{2}\overrightarrow{MQ}$, $\overrightarrow{RS} = \frac{1}{4}\overrightarrow{OR}$, oe $\overrightarrow{NS} = \frac{1}{2}\overrightarrow{MN}$, $\overrightarrow{MS} = \frac{3}{2}\overrightarrow{MN}$ $\overrightarrow{NS} = \frac{1}{3}\overrightarrow{MS}$